Scope	Global
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Classification	Main Document
Title	Validation of Computer Systems
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GENERAL INFORMATION

Applicability Impacted Division(s)	Animal Health, Biopharmaceuticals, BIX, Commercial - R/B-OPU, Development, Finance & Controlling, Global Business Services, Global Facilities & Engineering, Global Quality, Human Pharma Supply, Information Technology, Internal Auditing, Internal Project Management, Legal, Medicine, Purchasing, Research
Owning/Issuing Division	Global Quality
Signatories	See entry in Document Management System or print out

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1 PURPOSE

This SOP describes the requirements for the validation of Computer Systems. Computer systems include application systems and infrastructure systems.

Defines the general principles and detailed process steps for validation of computer systems following a System Life Cycle approach to meet both our external obligations to regulatory authorities as well as our own internal policies and procedures.

2 APPLICABILITY

This SOP is applicable:

All staff involved in validation activities defined in this SOP

Applicable to all business functions involved in

- GxP regulated activities
- other regulatory requirements in the context of medicinal products for human or veterinary use

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3 DEFINITIONS & ABBREVIATIONS

See 028-BIS-00516-RD01 “Roles and Definitions”.

Term	Definition / Explanation
e-Document Management System (DMS)	Electronic tool for life cycle management of documents.

Abbreviation	Description
N/A	N/A

Infographics	Description
	Summary – short description of the topic
	Explanation – additional information to the topic
	Attention – notice taken of something interesting or important
	Remember - what you should remember after reading

4 IMPLEMENTATION AND/OR PRE-REQUISITES

The implementation of the SOP updates will be done by reading the document in LOS (Learning One Source) as well as with trainings. Trainings are offered on regular basis, training material is already available. Additional training material is under elaboration and will also be accessible on the LeVA intranet homepage in future.

As described in Chapter 4 of the document 028-BIP-00516 the Corporate SOPs and their associated templates are intended to be the sole reference to requirements for CSV globally. The single exception to this requirement is in the event a requirement to have the SOPs translated to local language exists. For more details please refer to Chapter 4 of the document 028-BIP-00516.

5 ROLES & RESPONSIBILITIES

For the detailed description of the roles please see 028-BIP-00516 RD01 “Roles and Definitions”.

If a Business System Lead is assigned for a system this role takes over all responsibilities and signatures of the Business SMEs defined in this SOP. Additional signatures by the System / Process Owner may be assigned to this role e.g. in the role assignment log.

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The main responsibilities for the roles regarding validation activities are summarized in the following table:

Process Step		Responsibility					
		System Owner / Process Owner	Business SME or Business System Lead	Validation Manager	IT System Lead	IT SME	CSV&C Function
6.6	**General System Life Cycle Tasks**						
6.6	**Traceability**			C			
6.6.2	**Risk Management Process**	A	C	R		C	A
6.6.3	**21 CFR Part 11 Compliance Assessment**	A		C	C		RA
6.6.4	**Validation Documentation Storage**						X
6.6.5	**GLP Compliance Assessment**	A		C	C		RA
6.7	**The System Life Cycle**						
6.7.1	**Validation Planning, User Requirements, and Supplier Selection**						
6.7.1.1	**Validation Determination**	RA	C				A
6.7.1.2	**Validation Plan**	A	R	C/R			A
6.7.1.3	**Role Assignment Log**	RA		C			RA
6.7.1.4	**User Requirements Specification**	A	C	R			A
6.7.1.5	**Supplier Assessment and Supplier Selection**	X		X			X
6.7.1.6	**Cloud Service Provider**	A	C	R			RA
6.7.2	**Analysis and Design**						
6.7.2.1	**Functional Specifications**	A[1)]	C[2)] / A[1)]		R	C[2)]	
6.7.2.2	**Configuration Specifications**	A[1)]	C[2)] / A[1)]		R	C[2)]	
6.7.2.3	**System Design, Technical Installation, and Configuration & Installation Testing**			X	X		
6.7.2.4	**Software Design and Software Development**				X		
6.7.3	**Functional Testing and Requirements Testing**						

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Process Step		Responsibility					
		System Owner / Process Owner	Business SME or Business System Lead	Validation Manager	IT System Lead	IT SME	CSV&C Function
6.7.3.1	**Functional Testing / Requirements Testing Plan**		R	C/R			A
6.7.3.2	**Functional Test Cases and Requirements Test Cases**		RA	C/R			
6.7.3.3	**Signature Resource Log**			X			
6.7.3.4	**Execution of Testing**			X			
6.7.3.5	**Functional / Requirements Testing Report**		R	C/R			A
6.7.4	**Summary and Release into Operation**						
6.7.4.1	**Business Process SOPs**	X					
6.7.4.2	**Plans for Operation**			X			
6.7.4.3	**Installation in the Production Environment**			X	X		
6.7.4.4	**Data Migration**			X	X		
6.7.4.5	**Validation Summary Report**	A	R	C/R			A
6.7.5	**Client Site Qualification for Global Application Systems**			X			
6.7.6	**Operation and Maintenance of the System**			X	X		
6.7.7	**Retirement of the System**						

1) System / Process Owner or a delegate Business SME
2) Business SME or IT SME

C/R means the role typically creates the deliverable. If not the author the role reviews the deliverable.

Task	Abbreviation
Create	C
Review	R
Approve	A
E**X**ecute (no signature, but an activity)	X

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6 PROCESS

This section provides the general overview of the system life cycle activities and deliverables for the validation of a computer system. The following process step are described in detail:

- Validation Determination
- Validation Plan & Summary Report
- User Requirements / Functional / Configuration Specifications
- Functional / Requirements Testing
- Traceability
- Risk Management

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6.1 GENERAL PRINCIPLES

The System Life Cycle consists of activities that are organized into several steps and which are closely related to the Principles of Computer System Validation (CSV) within Boehringer Ingelheim.

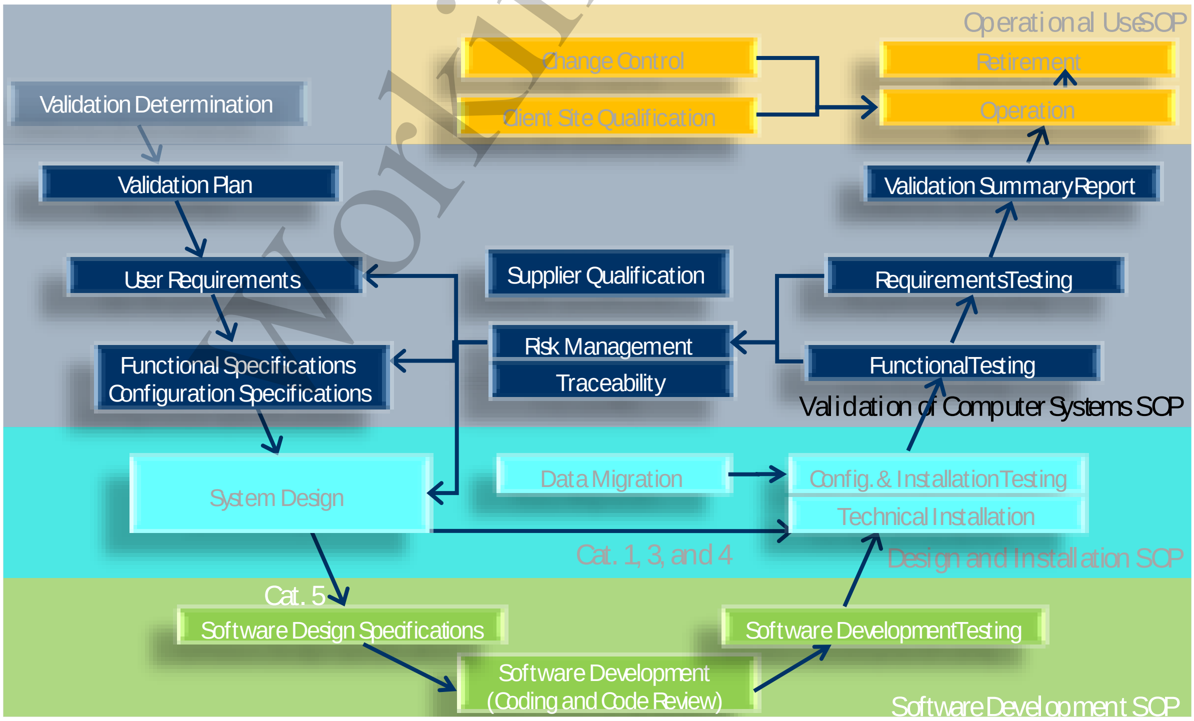

Fig. 1: Complete System Life Cycle for Computer Systems. Topics covered in referenced SOPs are shown in the light grey area

6.2 SYSTEM LIFE CYCLE FOR DIFFERENT GAMP CATEGORIES

GAMP 5 has defined different categories of software products for which different system life cycle activities and deliverables apply. For the purpose of this SOP the following four categories have to be considered:

GAMP Software Category	Description	Comment
1	This category comprises two kinds of software: • Layered software products required to develop or run applications (e.g. operating systems, database software, programming tools, middleware, spreadsheet software) • Infrastructure software tools (e.g. batch job scheduling tools, security software, anti-virus, configuration management tools).	Most application systems require this type of software as components to run the application software. Infrastructure systems only consist of category 1 software.
3	**Non-configured products:** Off-the shelf software that is either not configurable or used with the default configuration for the user specific business process. It is installed and used as provided by the supplier.	Examples: Commercially available software installed on standard hardware, Off-the-shelf software
4	**Configured products:** Off-the- shelf software that has to be configured for the user specific business process using standard interfaces and functionalities.	Any additional modification of the software codes such as application specific scripting languages shell scripts, batch files, customized reports and interfaces etc. have to be handled as Category 5
5	**Custom application:** Software is being developed to meet the user requirements for the user specific business process.	None

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The following table indicates which life cycle deliverables are typically required for the individual categories:

Deliverable / Activity	**Category 1[2)]**	**Category 3**	**Category 4**	**Category 5**
Validation Plan	X	X	X	X
User Requirements Specification	-	X	X	X
Functional Specification	X	-[3)]	-[3)]	X
Configuration Specification	-	-	X	(X)[1)]
Risk Assessment	X	X	X	X
System Design Specification	X	X	X	X
Software Design Specification	-	-[3)]	-[3)]	X
Source Code Review	-	-[3)]	-[3)]	X
Development Testing	-	-[3)]	-[3)]	X
Configuration & Installation Testing	X	X	X	X
Functional Testing	X	-	X	X
Requirements Testing	-	X	X	X
Validation Summary Report	X	X	X	X

[1)] if the system is based on a configurable software product
[2)] applicable for infrastructure systems only. Application systems always contain also GAMP category 3, 4 or 5 software
[3)] For GAMP category 3 and 4 these deliverable are under the responsibility of the software vendor. Existence should be verified during a vendor audit.

It should be taken into consideration that many systems will consist of a combination of several GAMP categories E.g. a configurable off-the shelf product (category 4) might be combined with a category 1 product MS Word, and custom programming (category 5) is done to create MS Word documents. In these cases the highest applicable category has to be applied to the validation package (i.e. for the given example category 5). In the validation plan, the details for the individual deliverables should be defined, for which components of the system they apply. For the above example the software design specification would be limited to the customization, but not include the category 1 and 4 products.

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6.3 INFRASTRUCTURE SYSTEMS

Infrastructure systems which are used by GxP relevant application systems have to be qualified rather than validated. Infrastructure systems only consist of software components of GAMP category 1 (see section 6.2). Systems comprising other software categories, i.e. non-configured products of GAMP category 3, configurable software (GAMP category 4) or any custom developed code (GAMP category 5) need to be validated as an application system.

However, for the sake of simplicity throughout this SOP and the related SOPs the term 'validation' is used instead of 'qualification'. Therefore, for infrastructure systems following replacements have to be made for the corresponding deliverables:

- Validation determination → Qualification determination
- Validation plan → Qualification plan
- Validation summary report → Qualification summary report.

For infrastructure systems the following SLC documents are not applicable:

- User requirements specifications
- Configuration specifications
- Requirements testing
- Business process SOPs
- Client Site Qualification
- 21 CFR Part 11 Compliance Assessment
- Retirement Determination

Typically a supplier assessment for infrastructure systems is not required because widespread industry standard hardware / software products are used.

Infrastructure systems are owned by the IT function. Consequently all roles and signatures for the validation of application systems are replaced by comparable IT roles and signatures, e.g.

- Business SME (Subject Matter Expert) → IT SME
- CSV&C Function → IT CSV&C Function
- System / Process Owner → System Owner

The review of functional testing plan / report is done by the *IT System Lead* or the *Validation Manager*.

For some deliverables separate templates exist for infrastructure systems. These templates are marked by the addition of '-I' in the template name and are also listed in 028-BIP-00516-RD02.

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6.4 CLOUD SYSTEMS (SAAS/PAAS/IAAS)

For cloud-based computer systems (SaaS/PaaS/IaaS) a significant part of the validation / qualification activities and associated deliverables are under the full control and responsibility of the supplier. Refer to 028-BIP-00516-RD03 for detailed guidance on the validation lifecycle approach.

The responsibilities for computer system in scope of the validation determines the templates to be used.
To facilitate validation planning and reporting of cloud-based computer systems the following dedicated templates are provided:

- Validation Plan: C-VPC
- Qualification Plan: C-QPC-I
- Validation Summary Report: C-VSRC
- Qualification Summary Report: C-QSRC-I

Please Note: In case an application system is installed under BI responsibilty on a qualified PaaS or IaaS, the templates C-VP, C-VSR should be used.

To support the supplier qualification and the risk-based approach a Cloud Service Provider Assessment should be performed as part of the validation. For details refer to section 6.7.1.6.

6.5 PREREQUISITES

For new systems, the implementation project has been initiated (not in the scope of this SOP). For existing validated systems a formal change control has been approved. *The System / Process Owner*, *Validation Manager*, *Computer Systems Validation and Compliance (CSV&C) Function*, and *Information Systems (IS) System Lead* and other *Subject Matter Experts (SMEs)* (as appropriate) have been assigned. Initial assignment of the roles is documented in the role assignment log (see section 6.7.1.3) and approved by the *System / Process Owner*. The role assignment log needs to be created and approved before any other life cycle document is approved. (see section 6.7.1.3).

6.6 GENERAL SYSTEM LIFE CYCLE TASKS

6.6.1 Traceability

A key task within the project and system life cycle is traceability. Traceability links the user requirements, risk assessment, functional specifications, design specifications, and all formal testing or mitigating factors (e.g. procedural controls) in a methodical and easy-to-follow manner.

It ensures that all requirements can be traced to a test or verification activity, where applicable, all requirements are met and can be traced to the appropriate functional specifications, where applicable, all functional specifications are met and can be traced to the appropriate design specifications.

The traceability may consist of a matrix or other means (such as cross-references within each document or via linkage in an electronic system). The template C-TRC must be used to create a traceability matrix in tabular format.

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The *Validation Manager* assures the correctness and completeness of the traceability in each step of the SLC.

6.6.2 Risk Management Process

Risk management is conducted throughout the lifecycle of a system until retirement. The following sections describe the elements of risk management along the system lifecycle.

6.6.2.1 Risk Management: Validation Planning

The *Validation Manager* together with all relevant roles (e.g. Business SMEs, System / Process Owner) analyses the overall criticality of the system considering the business process as well as regulatory obligations of a system as part of the validation plan (see 6.7.1.2). By signing the validation plan the *System / Process Owner* accepts the assessment of the overall criticality, the risk-based approach considerations, and all associated risks.

6.6.2.2 Risk Management: Risk Assessment

Based on the user requirements specifications, functional specifications (if applicable), and design specifications (if applicable) the *Validation Manager* and or *Business / IT SME* analyse the detailed criticality of individual requirements / specifications or groups of requirements / specifications considering criteria such as the business process, GxP and regulatory obligations. The risk assessment is documented using the template C-RA (for Infrastructure Systems: C-RA-I).

Any newly identified risks, or changes to any of the related specification documents during the system life cycle require an update of the risk assessment document.

If applicable, identify data fields relevant for the review of audit trail (see BI-VQD-09775-S, section Administration).

A *Business Subject Matter Expert (SME)* authors the risk assessment document with input by an IT SME.

The *Validation Manager* reviews the risk assessment document.

The *System / Process Owner* and the *CSV&C Function* approve the risk assessment document.

6.6.2.3 Risk Management: Risk-based Testing

The extent of formal testing will be based on the results of the risk assessment. Details on the risk-based testing approach have to be defined in the functional / requirements testing plan.

6.6.2.4 Risk Management: Final Risk Evaluation

The *Validation Manager* finally evaluates all the occurred risks considering failed test cases, missing documentation and SOPs, training issues, technical limitations, etc., in a risk assessment as part of the validation summary report (see section 6.7.4.5). The decision and justification to release the system to production must be documented. Risk mitigating measures need to be defined for unresolved failures or unacceptable risks.

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6.6.2.5 Risk Management: Periodic Review

The *Validation Manager* and the *CSV&C Function* determine if assumptions made for the overall criticality of the system and risk-based approach considerations are still valid as part of the periodic review of the system (see 6.7.4.5).

6.6.3 21 CFR Part 11 Compliance Assessment

As part of the validation plan (see 6.7.1.2) the *Validation Manager* together with all relevant roles (e.g. Business SMEs, CSV&C Function) determines if 21 CFR Part 11 and other regulations (e.g. EudraLex 11, 15) are applicable to the system.

If 21 CFR Part 11 is applicable to the system, the *Validation Manager* and the *IT System Lead* create an 21 CFR Part 11 compliance assessment using template C-P11.

The *CSV&C Function* reviews and approves the 21 CFR Part 11 compliance assessment.

The *System / Process Owner* approves the 21 CFR Part 11 compliance assessment.

The 21 CFR Part 11 compliance assessment needs to be completed and approved at latest before the system is released into production.

6.6.4 Validation Documentation Storage

The *CSV&C Function* has to verify that a process is in place for the storage and/or archiving of validation documents and all related documents (e.g. change control documentation, incident reports).

6.6.5 GLP Compliance Assessment

As part of the validation plan (see 6.7.1.2) the *Validation Manager* together with all relevant roles (e.g. Business SMEs, CSV&C Function) determines if GLP is applicable to the system.

If GLP is applicable to the system, the *Validation Manager* and the *IT System Lead* create an GLP compliance assessment using template C-GLP.

The *CSV&C Function* reviews and approves the GLP compliance assessment.

The *System / Process Owner* approves the GLP compliance assessment.

The GLP compliance assessment needs to be completed and approved at latest before the system is released into production.

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6.7 THE SYSTEM LIFE CYCLE

6.7.1 Validation Planning, User Requirements, and Supplier Selection

6.7.1.1 Validation Determination

If it is not clear if the system will be used in a GxP process or will affect GxP processes, a *Business SME* creates a validation determination using the template C-VD (for qualification determination of Infrastructure Systems: C-QD-I) to determine whether validation is required. This document provides a rationale for that decision based on the planned functionality of the system and the processes that will be supported by the system, and all applicable regulatory and legal requirements.
The *System / Process Owner* reviews and approves the validation determination. The *CSV&C Function* approves the validation determination.
In the qualification determination of an infrastructure system an assessment has to be made to identify whether the infrastructure system is used / interfaced by a validated GxP application system.
The review period for Validation Determinations is every 3 years in case a system is not validated. The review should include the verification of assigned Business SME and System Owner.
Within 3 years after approval of a validation determination the CSV&C Function initiates a review of this document, this can be documented with creation of a new Version of the document. A Business SME assesses and updates the document based on the actual functionality and processes supported by the System. If no changes are required the review is documented in the document history. The System / Process Owner reviews and approves the new Version of the Validation Determination. The CSV&C Function approves the validation determination. If the result of the review is that the System requires validation the System / Process Owner initiates a Validation Project.
In this case, a Business SME updates the Validation Determination, providing a link to the newly created Validation Plan and including a statement to explain, why the system will be validated going forward. The updated Validation Determination will be reviewed and approved by the System/Process Owner and approved by the CSV&C function.

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> A Validation Determination is only required if the system will not be validated. If the decision is to validate the system a Validation Plan is created instead to initiate the validation process.

6.7.1.2 Validation Plan

For systems to be validated a validation plan has to be created using the template C-VP or C-VPC (for qualification plan of Infrastructure Systems: C-QP-I or C-QPC-I). A single validation plan can also be used for a specific set of systems.

The *Validation Manager* and the *CSV&C Function* decide on modifications of the validation approach and validation deliverables based on the overall criticality of the system, applicable GAMP categories (see section 6.2) and other factors applicable to the system (i.e. risk based approach). All modifications must be documented and justified as part of the validation plan.

The *Validation Manager* should determine together with the *CSV&C Function* if SLC documentation (e.g., specifications, testing documentation) provided by the supplier can be leveraged for the validation approach. If supplier's SLC documentation will be leveraged (to replace BI internal deliverables), the *Validation Manager* must document this within the validation plan.

If an agile methodology will be used for the development of the system the *Validation Manager* and the *IT System Lead* define the agile approach including a reference to a related guidance document in the validation plan.

Checklists for the fulfilment of a specific predicate rule listed in 028- BIP-00516-RD02 "List of Templates" that are applicable for the computer system have to be identified within the Validation Plan e.g. checklist for 21 CFR 11 requirements (see section 6.6.3).

Typically the *Validation Manager* creates the validation plan. If the validation plan is created by another *Business SME* the *Validation Manager* reviews the validation plan.

A *Business SME* reviews the validation plan.

The *System / Process Owner* approves the validation plan. By approving the validation plan the *System / Process Owner* also accepts the specific risk based approach to the validation of the computer system.

The *CSV&C Function* approves the validation plan. This approval includes the review for compliance with applicable GxP rules and regulations.

6.7.1.3 Role Assignment Log

The *Validation Manager* creates a roles assignment log using template C-RAL. The *Validation Manager* is responsible to keep this document up-to-date until the system has been retired. The *System / Process Owner* and the CSV&C Function review and approve the role assignment log.

6.7.1.4 User Requirements Specification

A *Business SME prepares* the user requirements specification using template C-URS.

The *Validation Manager* reviews the user requirements specification.

The *System / Process Owner* and the *CSV&C Function* approve the user requirements specification.

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6.7.1.5 Supplier Assessment and Supplier Selection

For supplier qualification of software / service providers the existing processes / SOPs of the affected business area for supplier qualification and oversight shall be followed.

The *System / Process Owner* is responsible for ensuring that supplier of computer systems and/or any service providers will meet technical, commercial, and regulatory requirements (e.g. through an supplier assessment). The assessment may be performed by mail assessment (using template C-SMQ for Software providers or C-CSPQ for Cloud Service providers), as an onsite audit or assessment based on the following risk based approach.

6.7.1.6 Cloud Service Provider Assessment

For cloud services (SaaS/PaaS/IaaS) the supplier shall complete a cloud service provider questionnaire (C-CSPQ) to provide information related to validation and compliance of the services provided. The *Validation Manager* archives the completed questionnaire as part of the validation documentation.

The *Business System Lead* (for SaaS) or the *IT System Lead* (for PaaS/IaaS) completes a cloud service provider assessment using template C-CSPA.

The *CSV&C Function* reviews and approves the cloud service provider assessment.

The *System / Process Owner approves* the cloud service provider assessment.

The *Validation Manager* is responsible to consider in the validation plan actions and mitigating measures which have been identified in the assessment.

6.7.2 Analysis and Design

6.7.2.1 Functional Specifications

Based on the user requirements specifications the functional specifications have to be created using template C-FS (for Infrastructure Systems: C-FS-I). Functional specifications must be prepared for all developed systems (GAMP Cat. 5 system) or customizations of commercial off-the shelf systems (GAMP Cat. 4 system). The functional specifications must also clearly address user requirements that will, and those that will not, be fulfilled.

A *Business SME* or an *IT SME* creates the functional specifications.

The *IT System Lead* reviews the functional specifications.

The *System / Process Owner* or a delegate *Business SME* approves the functional specifications.

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6.7.2.2 Configuration Specifications

Based on the user requirements specifications the configuration specifications have to be created using template C-CS. Configuration specifications must be prepared for GAMP Category 4 and 5 systems (if based on configurable software product).

The configuration specifications must contain and define all configurable items of the system required to adapt the system to the specific business processes and / or required functionalities according to the user requirement specification and / or functional specifications. Technical configuration settings to adapt the system to the technical environment are handled under 6.7.2.3 System Design, Technical Installation, and Configuration & Installation Testing .

A *Business SME* or an *IT SME* creates the configuration specifications.

The *IT System Lead* reviews the configuration specifications.

The *System / Process Owner* or a delegate *Business SME* approves the configuration specifications.

6.7.2.3 System Design, Technical Installation, and Configuration & Installation Testing

The *IT System Lead* is responsible to ensure that system design, technical installation, and configuration & installation testing of the system are performed according to the validation plan and the SOP BI-VQD-08127-S "Computer System Design and Installation".

The *Validation Manager* verifies the existence of the configuration & installation testing report for the validation environment prior to start of functional testing and/or requirements testing. All changes to the qualified validation environment must be performed either under an ongoing initial validation or an existing change control.

In case of changes to the validation environment only a change control is required.

6.7.2.4 Software Design and Software Development

The *IT System Lead* is responsible to ensure that software development of the system or customizations of the system are performed according to the validation plan and the SOP BI-VQD-09852-S "Computer System Software Development".

6.7.3 Functional Testing and Requirements Testing

6.7.3.1 Functional Testing / Requirements Testing Plan

Typically the *Validation Manager* prepares a functional / requirements testing plan using the template C-FTP (for Infrastructure Systems: C-FTP-I). If created by a *Business SME* the *Validation Manager* reviews the functional / requirements testing plan.

A *Business SME* reviews the functional / requirements testing plan.

The *CSV&C Function* approves the functional / requirements testing plan.

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6.7.3.2 Functional Test Cases and Requirements Test Cases

Typically the *Validation Manager* prepares the functional test cases and / or requirements test cases as specified in the functional / requirements testing plan using template C-TC. If created by a *Business SME*, the *Validation Manager* reviews and approves the test cases.

Functional test cases must be traced to functional specifications and / or configuration specifications. Requirement test cases must be traced to user requirement specifications and risk mitigations identified in the risk assessment.
A *Business SME* reviews and approves the functional and/or requirements test cases.

Test cases may be attached to the functional testing / requirements testing plan without separate signature.

6.7.3.3 Signature Resource Log

The *Validation Manager* creates a signature resource log using the template C-SRL. This log will contain a handwritten signature and initials of each person who will sign any test documentation.

A signature resource log is not required if test results are documented in an electronic system without any handwritten signatures or initials.

6.7.3.4 Execution of Testing

The functional / requirements testing plan must be approved prior to starting any testing activities. Individual test cases must be approved before execution.

The *Validation Manager* ensures that the testers are trained and their training is documented. The *Validation Manager* supervises the testing and reviews the executed test cases. For failed test cases the *Validation Manager* provides appropriate comments directing the resolution.

Test results and supporting evidence need to be recorded immediately and accurately following Good Documentation Practices, and manually signed/initialled and dated by the tester. If test results are recorded using an electronic system the system must support Good Documentation Practices and must be validated.

For test case failures that cannot be resolved during execution of the test case (i.e. test case failed by a second reviewer), the *Validation Manager* completes the discrepancy log using the template C-DIL.

6.7.3.5 Functional / Requirements Testing Report

Summary and evaluation of functional testing and / or requirements testing results may be done in a separate report or as part of the validation summary report (refer to section 6.7.4.5). Functional and requirements testing may be reported in a combined report, or in two separate reports. The detailed reporting process must be defined in the validation plan.

If separate reports are to be created typically the *Validation Manager* creates a functional / requirements testing report using template C-FTR (for Infrastructure Systems: C-FTR-I). If

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created by a *Business SME*, the *Validation Manager* reviews the functional / requirements testing report.

A *Business SME* reviews the functional / requirements testing report.

The *CSV&C Function* approves the functional / requirements testing report.

6.7.4 Summary and Release into Operation

6.7.4.1 Business Process SOPs

The *System / Process Owner* makes sure that the business process SOPs are prepared or modified, as specified in the validation plan. The existence and effective state of new or updated SOPs should be verified in the validation summary report. The business process SOPs must be effective when the system is released into operation phase.

6.7.4.2 Plans for Operation

The *Validation Manager* ensures that the plans for operation of the system are effective before the system is released for operation, as specified in the validation plan and the SOP BI-VQD-09775-S "Operational Use of Computer Systems".

6.7.4.3 Installation in the Production Environment

If the system requires installation into the production environment (meaning the validation and production environments are different) the technical installation, configuration, and installation qualification is done according to SOP BI-VQD-08127-S "Computer System Design and Installation".

The *IT System Lead* is responsible to ensure that the configuration and installation testing plan and report for the production environment are prepared per SOP BI-VQD-08127-S "Computer System Design and Installation".

The *Validation Manager* verifies the existence of the configuration and installation testing report for the production environment prior to release of the system into operation.

6.7.4.4 Data Migration

If data needs to be migrated (e.g. imported from a legacy system into the new system) the *IT System Lead* is responsible to ensure that a data migration plan according to SOP BI-VQD-08127-S "Computer System Design and Installation" is prepared. The data migration plan describes how data will be transferred, which tools are used and how the migrated data will be checked for accuracy, consistency and completeness.

Upon completion of the data migration the *IT System Lead* is responsible to ensure that a data migration report according to SOP BI-VQD-08127-S "Computer System Design and Installation" is prepared.

The *Validation Manager* verifies the existence of the data migration report for the production environment prior to release of the system into operation.

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6.7.4.5 Validation Summary Report

The validation summary report has to be created using template C-VSR or C-VSRC (for Infrastructure Systems: C-QSR-I or C-QSRC-I). It must contain a statement whether the system can be released for operation in a regulated environment. It also defines the period to review the validation status according to the SOP BI-VQD-09775-S "Operational Use of Computer Systems". The maximum time to perform a periodic review for a system is **three years**. The period should be defined based on overall system risk, complexity, and expected number of changes per year and should be justified.

If any deviations have occurred during the validation project (e.g. deviations from the validation plan or deviations from a mandatory template), the *Validation Manager* must list the deviations within the validation summary report and provide an assessment for their acceptance and potential impact on the validated state of the system.

Typically the *Validation Manager* creates the validation summary report. If created by a *Business SME*, the *Validation Manager* reviews the validation summary report.

A *Business SME* reviews the validation summary report.

The *System / Process Owner* approves the validation summary report. By approving the validation summary report the *System / Process Owner* also accepts the deviations and corrective measures defined in the validation summary report.

The *CSV&C Function* approves the validation summary report. This approval includes the review for compliance with applicable GxP rules and regulations.

6.7.5 Client Site Qualification for Global Application Systems

If an application system is installed centrally, but used at more than one site or OPU a client site qualification should be executed for each site before the site gets access to the production system. The *Validation Manager* initiates the client site qualification process as defined in the SOP BI-VQD-09775-S "Operational Use of Computer Systems".

The validation plan should define if a client site qualification is required, and for which initial sites or OPUs a client site qualification must be executed prior to release of the system into operation. The client site qualification process after production release has to be detailed in the Application Administration Plan.

6.7.6 Operation and Maintenance of the System

In order to maintain the validated state of the system the *Validation Manager* and *IT System Lead* are responsible to ensure that all plans for operation of the system as well as the processes defined in the SOP BI-VQD-09775-S "Operational Use of Computer Systems" are being followed.

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6.7.7 Retirement of the System

Retirement is the formal process for reclassifying the status of a validated computer system to denote that it will no longer support GxP relevant processes or data. Therefore, retirement shall be applied when such a system is being withdrawn from operation entirely or when such a system is to remain in operation but will no longer support GxP relevant activities. The retirement process is defined in the SOP BI-VQD-09775-S "Operational Use of Computer Systems".

6.8 TEMPLATES FOR VALIDATION DOCUMENTATION

Templates for validation deliverables are available (refer to 028-BIP-00516-RD02 "List of Templates for CSV Documentation"). While the format of these templates can be modified (e.g. letter size), the content is binding, including structure and sequence. Electronic systems may be used as a replacement of mandatory templates provided that the mandatory content of the template is kept by the electronic system, and that the system has been validated.

The templates (could) consist of four different kinds of text/sections (the different text / sections will be marked with different colors within the templates

a) Mandatory text / sections
b) Recommended text / sections
c) Text / sections to be chosen from (e.g. two alternatives descriptions available)
d) Instructions to guide / support the authors (have to be deleted before the „document“ is finalized)

If supplier provided SLC documentation is found to be adequate and meets the requirements of BI internal SOPs and the associated templates, it may be used instead of the BI internal templates.

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7 ASSOCIATED DOCUMENTS

No.	Name	Title
1	N/A	N/A

8 REFERENCES

No.	Name	Title
1	028-BIP-00516	Principles of Computerized Systems Validation
2	028-BIP-00516-RD01	Roles and Definitions
3	028-BIP-00516-RD02	List of Templates for CSV Documentation
4	028-BIP-00516-RD03	Internal Requirements for Use of Cloud-Computing for GxP Applications
5	BI-VQD-08127-S	Computer System Design and Installation
6	BI-VQD-09852-S	Computer Systems Software Development
7	BI-VQD-09775-S	Operational Use of Computer Systems

9 DOCUMENT HISTORY

Version	Description of Changes	Author	Expert team
6.0	Adaption to new template for SOPs Adaption of referenced SOP numbers 6.4: Added new section for Cloud systems 6.6.5: Added section C-GLP Compliance Assessment 6.7.1.1: Added review of Validation Determination documents every 3 years 6.7.1.2 Removed statement for Global Master Data Governance 6.7.1.5: Section updated and added C-CSPQ 6.7.1.6: New section for Cloud Service Provider Assessment 6.7.2.3: Added requirement for Change control in case of changes to a validation environment 6.7.4.5: Added templates for cloud and updated wording.	Myriam Wolter, Global Quality IT Michael Eisele, Global Quality IT	Members of the ICSVC Working Group

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Document Approvals

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